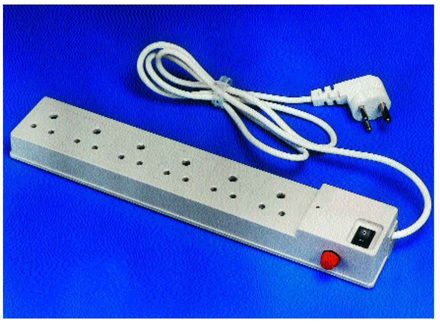


■ **Stabiliser:** A stabiliser is a slightly better device for power protection—it is capable of detecting both power surges and brownouts and can rectify them to deliver a regulated AC voltage to your computer. It continuously senses the input voltage (and in some cases even the output voltage). It uses a closed feedback circuitry to clamp the output voltage to a constant value by either 'bucking' (lowering) or 'boosting' (increasing) the input voltage, depending on its level.

The principal element of a stabiliser is a transformer whose output voltage is controlled depending upon the input voltage. It will control the output voltage between the low and high voltage thresholds at which the device operates, but if these limits are surpassed, it simply switches off. A stabiliser cannot regulate the frequency of the output voltage or provide any form of backup power, but it does provide adequate control of voltage for most applications in case of occasional low-level fluctuations in power.

Choosing the Right UPS Rating

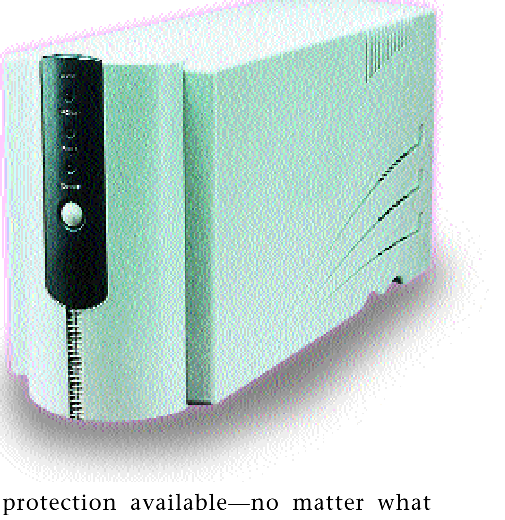
Nowadays, almost every piece of electronic equipment you buy specifies the amount of current it draws in amperes (A). Multiply this by the voltage we use (230 volts) and you get the VA rating. If the power rating is specified in watts, simply multiply this by a factor of 1.4 and you'll get the VA rating. This is the amount of power the device draws. Now, if you simply add up the individual VA ratings of all the devices you plan to connect to your UPS and add a 10 per cent safety factor, you will be able to decide the rating of the UPS you need.

As an example, the average current drawn by a desktop computer today would be: 1.2 A for a 17-inch monitor and 1.1 A for a 1.4 GHz processor-based computer. After multiplying each of these by 230 volts, we will have a power rating totalling: 276 + 253 = 529 VA. This is a maximum-limit scenario considering that you will have a fully loaded computer system. Therefore, to be on the safe side, buy at least a 600 VA system to run your computer. To alleviate the load on your UPS, do not connect devices such as printers, scanners, speakers, etc.

■ **UPS:** This is the most effective way to implement power protection. A UPS is a device capable of maintaining pure AC power and can even provide backup power in case of very poor input power or a power failure. Here, DC power provided by the UPS' batteries is converted into AC power by an electronic circuit called an inverter. This gives clean power to the equipment connected to it.

There are three types of UPS systems in use—offline, on-line and line-interactive. In an offline UPS, the equipment is driven directly by the mains power as long as it remains within threshold limits of the minimum and maximum values. During this time, the batteries of the UPS are continually charged. In case of an extreme power condition or a power failure, the UPS switches over very quickly (usually under 5 milliseconds) to the batteries.

In an online UPS, the equipment is continually driven through the batteries even during normal conditions, during which time the batteries also charge. Therefore, this type of UPS understandably provides the purest form of power protection available—no matter what the input power conditions, the output power is provided through the battery.



A line-interactive UPS system is a hybrid between the two preceding types—it incorporates a transformer that controls the output voltage within a specified range and allows the batteries to kick in only if this range is exceeded. This method overcomes the need for the UPS to switch over to the battery every time even a slight voltage fluctuation occurs, thereby extending its lifespan.